



EXAM PAPERS PRACTICE

Boost your performance and confidence with these topic-based exam questions

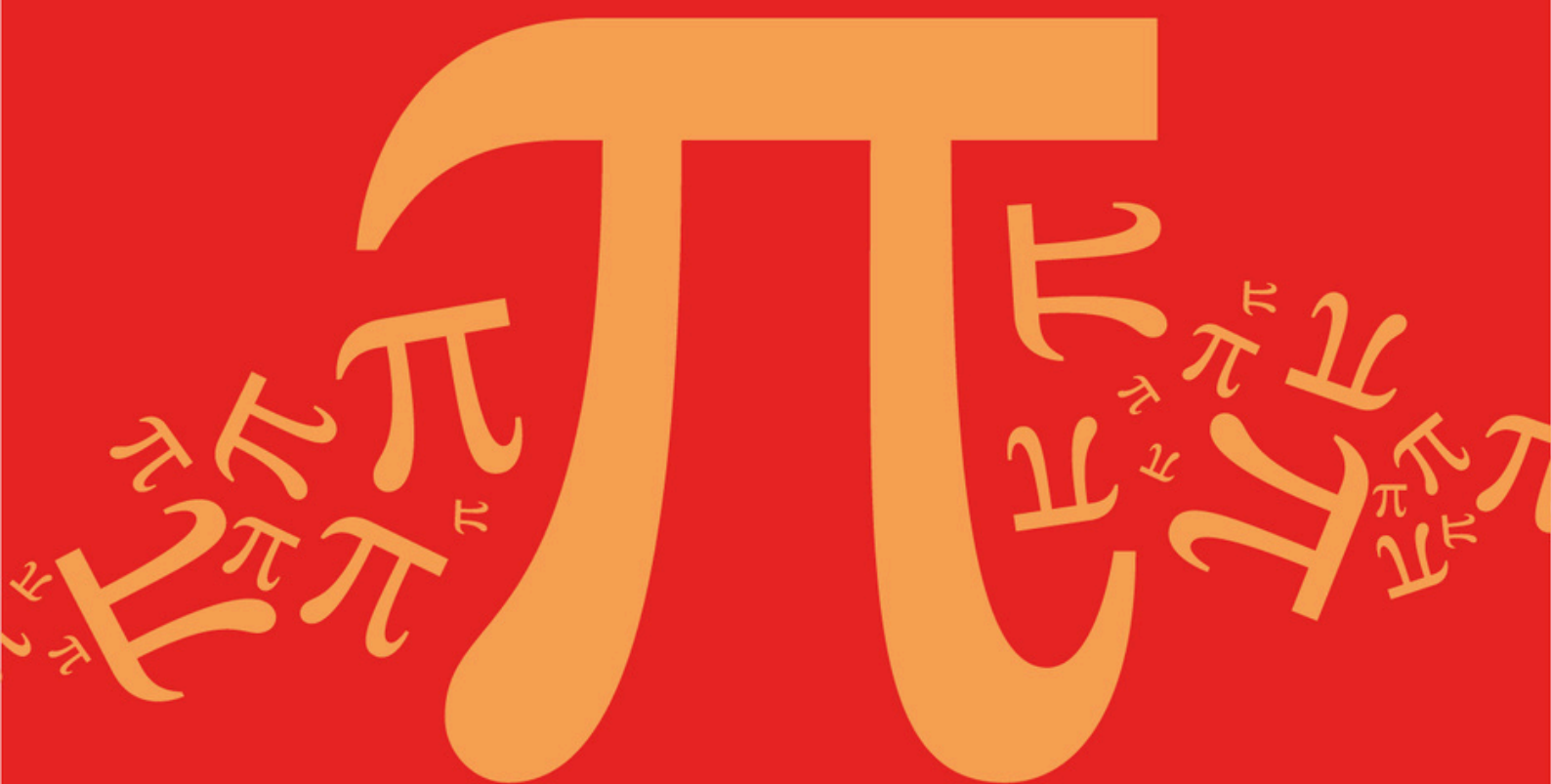
Practice questions created by actual examiners and assessment experts

Detailed mark scheme

Suitable for all boards

Designed to test your ability and thoroughly prepare you

AQA A Level Further Mathematics (7367)



Topic

D: Further Algebra and Functions

Sub topic

D6: Use Standard Maclaurin Series

Mark Scheme

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Mark Scheme

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**To be used by all students and teachers for
AQA A Level Further Mathematics (7367)
for exams from 2025 onwards.**

Students and teachers of other boards may also find this useful

Mark schemes

Q1.

Marking Instructions	AO	Marks	Typical Solution
Circles correct answer	AO1.2	B1	$\cos x$
Total 1 mark			

Q2.

Ignore higher powers beyond x^2 throughout this question

(a) $e^{3x} = 1 + 3x + 4.5x^2$

B1

1

(b) $(1 + 2x)^{-3/2} = 1 - 3x + \frac{15}{2}x^2$

$(1 + 2x)^{-3/2} = 1 \pm 3x + kx^2$ or $1 + kx \pm 7.5x^2$ OE

M1

$1 - 3x + 7.5x^2$ OE (simplified PI)

A1

$e^{3x} (1 + 2x)^{-3/2} = (1 + 3x + 4.5x^2)(1 - 3x + 7.5x^2)$

Product of c's two expansions with an attempt to multiply out to find x^2 term

M1

x^2 term(s): $7.5x^2 - 9x^2 + 4.5x^2 = 3x^2$.

A1

4

[5]

Q3.

(a) (i) $\cos x + \sin x = 1 + x - \frac{1}{2}x^2 - \frac{1}{6}x^3$

Accept coeffs unsimplified, even 3! for 6.

B1

1

(ii) $\ln(1 + 3x) = 3x - \frac{1}{2}(3x)^2 + \frac{1}{3}(3x)^3 = 3x - \frac{9}{2}x^2 + 9x^3$

Accept coeffs unsimplified

B1

1

(b) (i) $y = e^{\tan x}, \quad \frac{dy}{dx} = \sec^2 x e^{\tan x}$

Chain rule

M1

ACF eg $y \sec^2 x$

A1

$$\frac{d^2 y}{dx^2} = 2 \sec^2 x \tan x e^{\tan x} + \sec^4 x e^{\tan x}$$

Product rule OE

m1

ACF

A1

$$= \sec^2 x e^{\tan x} (2 \tan x + \sec^2 x)$$

$$= \frac{dy}{dx} (2 \tan x + 1 + \tan^2 x)$$

$$\frac{d^2 y}{dx^2} = (1 + \tan x)^2 \frac{dy}{dx}$$

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AG Completion; CSO any valid method.

A1

5

(ii) $\frac{d^3 y}{dx^3} = 2(1 + \tan x) \sec^2 x \frac{dy}{dx} + (1 + \tan x)^2 \frac{d^2 y}{dx^2}$

M1

When $x = 0$, $\frac{d^3 y}{dx^3} = 2(1)(1)(1) + (1)(1) = 3$
CSO

A1

2

(iii) $y(0) = 1; y'(0) = 1; y''(0) = 1; y'''(0) = 3;$

$$y(x) \approx y(0) + xy'(0) + \frac{1}{2}x^2 y''(0) + \frac{1}{3!}x^3 y'''(0)$$

M1

$$e^{\tan x} \approx 1 + x + \frac{1}{2}x^2 + \frac{1}{2}x^3$$

CSO AG

A1

2

(c) $\lim_{x \rightarrow 0} \left[\frac{e^{\tan x} - (\cos x + \sin x)}{x \ln(1 + 3x)} \right]$

$$= \lim_{x \rightarrow 0} \frac{1 + x + \frac{x^2}{2} + \frac{x^3}{2} - 1 - x + \frac{x^2}{2} + \frac{x^3}{6}}{x \left(3x - \frac{9}{2}x^2 + \dots \right)}$$

Using series expns.

$$= \lim_{x \rightarrow 0} \left[\frac{x^2 + \frac{2}{3}x^3 + \dots}{3x^2 - \frac{9}{2}x^3 \dots} \right] = \lim_{x \rightarrow 0} \left[\frac{1 + \frac{2}{3}x + \dots}{3 - \frac{9}{2}x \dots} \right]$$

Dividing numerator and denominator by x^2
to get constant terms. OE following a slip.

M1

m1

$$= \frac{1}{3}$$

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A1

3

[14]

Q4.

(a) (i) $e^{2x} = 1 + 2x + 2x^2 + \frac{4}{3}x^3 + \dots$

Clear use of $x \rightarrow 2x$ in
expansion of e^x

M1

ACF

A1

2

(ii) $\{f(x)\} = e^{2x}(1 + 3x)^{-\frac{2}{3}}$

$$(1+3x)^{-\frac{2}{3}} = 1 + \left(-\frac{2}{3}\right)(3x) + \frac{\left(-\frac{2}{3}\right)\left(-\frac{5}{3}\right)(3x)^2}{2} - \frac{40}{3}x^3$$

First three terms as

$$1 + \left(-\frac{2}{3}\right)(3x) + kx^2 \quad \text{OE}$$

M1

$$= 1 - 2x + 5x^2 - \frac{40}{3}x^3$$

A1

$\{f(x) \approx\}$

$$1 + 2x + 2x^2 + \frac{4x^3}{3} - 2x - 4x^2 - 4x^3 + 5x^2 + 10x^3 - \frac{40x^3}{3}$$

Dep on both prev MS

m1

Condone one sign or numerical slip in mult.

A1Ft

$$= 1 + 3x^2 - 6x^3$$

CSO AG A0 if

binominal series not used

A1

5

(b) (i) $y = \ln(1 + 2 \sin x) \Rightarrow \frac{dy}{dx} = \frac{1}{1 + 2 \sin x} - 2 \cos x$

Chain rule

M1A1

$$\frac{d^2y}{dx^2} = \frac{(1 + 2 \sin x)(-2 \sin x) - 2 \cos x(2 \cos x)}{(1 + 2 \sin x)^2} = \frac{-2(\sin x + 2)}{(1 + 2 \sin x)^2}$$

Quotient rule OE with
u and v non constant

M1

ACF

A1

4

(ii) $y(0) = 0, \quad y'(0) = 2, \quad y''(0) = -4$

M1

McL Thm: $\{\ln(1 + 2 \sin x)\} \approx 0 + 2x - 4\left(\frac{x^2}{2}\right) + \dots \approx 2x - 2x^2$
 CSO AG

A1

2

(c)
$$\lim_{x \rightarrow 0} \frac{1 - f(x)}{x \ln(1 + 2 \sin x)} = \lim_{x \rightarrow 0} \frac{-3x^2 + 6x^3}{2x^2 - 2x^3}$$

Using expansions

M1

$$= \lim_{x \rightarrow 0} \frac{-3 + 6x}{2 - 2x}$$

*Division by x^2 stage
before taking limit.*

m1

$$= -\frac{3}{2}$$

CSO

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Q5.

(a) (i) $f(x) = \ln(1 + e^x)$:

$$f(0) = \ln 2$$

B1

$$f'(x) = \frac{e^x}{1 + e^x} \quad f'(0) = \frac{1}{2}$$

Chain rule

M1

A1

$$f''(x) = \frac{(1 + e^x)e^x - e^x e^x}{(1 + e^x)^2} = \frac{e^x}{(1 + e^x)^2}$$

Quotient rule OE

M1

A1

$$f''(0) = \frac{1}{4}$$

so first three terms are:

$$f(x) = \ln 2 + \frac{1}{2}x + \frac{1}{4} \frac{x^2}{2!} = \ln 2 + \frac{1}{2}x + \frac{1}{8}x^2$$

CSO; AG

A1

6

(ii)

$$f'''(x) = \frac{(1+e^x)^2 e^x - e^x [2(1+e^x)e^x]}{(1+e^x)^4}$$

Chain rule with quotient/product rule
 ft on $f'(x) = ke^x (1+e^x)^n$ (integer $n < 0$)

M1
A1ft

$$f'''(0) = \frac{4-4}{2^4} = 0$$

CSO; AG;

A1

{so coefficient of x^3 is zero}

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3

SC for those not using Maclaurin's theorem: **maximum** of 4/9

(b)

$$\frac{1}{2}x + \frac{1}{8}x^2$$

B1

1

(c)

$$\ln\left(1 - \frac{x}{2}\right) = \left(-\frac{x}{2}\right) - \frac{1}{2}\left(-\frac{x}{2}\right)^2 + \frac{1}{3}\left(-\frac{x}{2}\right)^3 - \dots$$

B1

1

(d)

$$\ln\left(\frac{1+e^x}{2}\right) + \ln\left(1 - \frac{x}{2}\right) = -\frac{x^3}{24} + \dots$$

Uses previous expansions to obtain first
non-zero term of the form kx^3

M1

$$x - \sin x \approx x - \left[x - \frac{x^3}{3!} + \dots \right] \approx \frac{x^3}{3!} + \dots$$

B1

$$\left[\frac{\ln\left(\frac{1+e^x}{2}\right) + \ln\left(1-\frac{x}{2}\right)}{x - \sin x} \right] = \frac{-\frac{1}{24}x^3 + \dots}{\frac{1}{6}x^3 + o(x^5)}$$

M1

$$= \frac{-\frac{1}{24}x^3 + \dots}{x^3 \left[\frac{1}{6} + o(x^2) \right]} = \frac{-\frac{1}{24} + \dots}{\frac{1}{6} + o(x^2)}$$

$$\lim_{x \rightarrow 0} \dots = -\frac{1}{4}$$

CSO

A1

4

[15]

Examiner reports

Q1.

The answer of $\cos x$ was correctly chosen by most students, with the wrong choices evenly distributed amongst the other three options.

Q2.

Almost all students were able to write down the correct expansion of e^{3x} . In part (b), the great majority of students used the 'hence' approach in which the most common error was to obtain a wrong coefficient of the x^2 term, normally $\frac{15}{8}$ but more surprisingly, $\frac{25}{2}$, in the binomial expansion. Only a small number of students used the alternative method of differentiation of $e^{3x} (1 + 2x)^{-1.5}$ and applying Maclaurin's theorem.

Q3.

Most candidates were able to write down the required two expansions in parts (a).

In part (b)(i), candidates generally used the chain rule to find $\frac{dy}{dx}$ and then applied the product rule to find $\frac{d^2y}{dx^2}$. Although a common error was to differentiate $\sec^2 x$ as

$2\sec x \tan x$, those who did find $\frac{d^2y}{dx^2}$ correctly often produced a convincing solution to reach the printed answer.

Some very good solutions were seen for parts (b)(ii) and (b)(iii) but it was disturbing to see some other candidates make no attempt to find the third derivative in part (b)(ii), yet claim that $f'''(0) = 3$ and write down the printed result in part (b)(iii). Clearly in such cases marks cannot be awarded.

Candidates who attempted part (c) generally showed a thorough understanding of the process required, including the need for explicitly reaching the stage of a constant term in both the numerator and denominator before taking the limit as $x \rightarrow 0$.

Q4.

Most candidates were able to write down the first four terms in the series expansion of e^{2x} , but in part (a)(ii) a significant minority of candidates ignored the request to use the binomial series expansion and instead resorted to differentiation of $f(x)$ and the use of Maclaurin's theorem. In part (b)(i), the most common error was to fail to use the chain rule

and thus to write the derivative of $\ln(1 + 2 \sin x)$ as $\frac{1}{1 + 2 \sin x}$.

Those candidates who correctly applied the chain rule usually went on to find the correct

expression for $\frac{d^2y}{dx^2}$ by applying either the quotient rule or the product rule. Almost all

candidates who obtained an expression for $\frac{d^2y}{dx^2}$ showed a correct understanding of Maclaurin's theorem in part (b)(ii).

Although the majority of candidates could correctly find the limiting value of the given expression as x tended to 0 in part (c), there were some incorrect solutions seen which just involved substituting $x = 0$ into the given expression.

Q5.

Many candidates, including those who had scored almost full marks on the previous five questions, failed to score many marks on this question. In part (a), a minority of candidates attempted to use series expansions instead of the prescribed Maclaurin's theorem. Although a special case was allowed, with a significant reduction in available marks, very few of these candidates scored any of them. For those using Maclaurin's theorem, most scored the first mark for finding $f(0)$ but a large number of candidates failed

to use the chain rule and differentiated $\ln(1 + e^x)$ incorrectly as $\frac{1}{1+e^x}$. Some of these candidates 'recovered' to score the next method mark but others just continued incorrectly to give subsequent derivatives in the form $(1 + e^x)^n$. Part (b) was not answered well, the

most common error being illustrated by

$$\ln\left(\frac{1+e^x}{2}\right) = \ln\frac{2}{2} + \frac{1}{2}\left(\frac{1}{2}x\right) + \frac{1}{2}\left(\frac{1}{8}x^2\right) = \frac{1}{4}x + \frac{1}{16}x^2$$

Most candidates gave the correct answer for part (c) and in part (d) a significant majority of candidates scored at least half the marks. Some candidates failed to use sufficient terms in the series expansion of $\sin x$.